

Ensemble Methods: Boosting vs. Bagging

Final Project Report

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Abstract—This paper presents an empirical comparison of two ensemble learning paradigms—boosting and bagging—using decision-tree base learners. We implement a Decision Tree classifier, AdaBoost (with decision stumps), and a Random Forest from scratch. Controlled experiments on real-world datasets investigate scaling behaviour, noise robustness, and bias-variance decomposition. Unsupervised analysis (PCA, K-Means, DBSCAN) supplements the visual exploration of the data. Our findings show that boosting outperforms bagging on small, clean, binary problems (a +0.53% accuracy edge on Breast Cancer), while bagging scales better as data complexity grows, winning on Adult Income (+0.47%), MNIST 3-vs-8 (+2.99%), and decisively on the imbalanced, multi-class Covertypes dataset (+13.62% macro F1). AdaBoost is also markedly more sensitive to label noise than Random Forest.

Deliverables: In addition to this report, we provide the complete source code (tagged `v1.0-final`), a professional slide deck for the oral defense, and a separate contribution report detailing each team member’s specific work.

I. INTRODUCTION

The central question of this project is: “*Under what conditions does boosting outperform bagging, and vice versa, and why?*” To answer it, we implement the core algorithms from scratch and evaluate them on datasets of varying characteristics.

The remainder of the paper is structured as follows. Section II reviews essential literature. Section III describes our implementations. Section IV details the experimental setup and Section V presents the results. Section VI discusses the findings, and Section VII concludes. The accompanying slide deck and contribution

report are submitted separately as required by the project brief.

II. RELATED WORK

A. Decision Trees

Decision trees recursively partition the feature space to minimise impurity [1]. CART [2] uses Gini impurity or entropy.

B. Boosting

AdaBoost [3] iteratively re-weights samples, giving higher importance to misclassified instances. The weight update rule is $\alpha_m = \ln \frac{1-\epsilon_m}{\epsilon_m}$ for binary classification.

C. Bagging and Random Forests

Bagging [4] reduces variance by training bootstrap replicates. Random Forests [5] add feature sub-sampling to decorrelate trees further.

D. Unsupervised Techniques

PCA [6] finds directions of maximum variance. K-Means [7] partitions data into k clusters, while DBSCAN [8] discovers clusters of arbitrary shape based on density.

III. METHODS

We implemented all algorithms in Python, using only NumPy (and `sklearn.metrics` for ARI). The code is available at <https://github.com/Rufet132/v1.0-final> (tag `v1.0-final`).

A. Decision Tree

We follow the CART methodology with exhaustive split search. Both Gini impurity and entropy are supported, with `sample_weight` for the boosting module.

B. AdaBoost

The discrete SAMME variant [1] uses depth-1 stumps. Sample weights are updated each round and the stump is trained on weighted data.

C. Random Forest

Bootstrap samples are drawn with replacement. Trees are grown with feature sub-sampling (`sqrt` by default). OOB error is computed for each sample using trees where it was not in the bootstrap.

D. Unsupervised Pipeline

- **PCA:** Eigen-decomposition of the covariance matrix.
- **K-Means:** Lloyd’s algorithm with multiple restarts.
- **DBSCAN:** Brute-force region queries, expanding core points.

IV. EXPERIMENTAL SETUP

A. Datasets

We use four datasets (Table I).

B. Preprocessing

Features were standardised (zero mean, unit variance). Missing values were dropped/imputed (justify). Adult rows containing ‘?’ were dropped (< 8% of rows); categorical features were one-hot encoded; the scaler was fit on the training split only. For the imbalanced Covertypes dataset (minority class 0.47%, below the 1% threshold), we used balanced class weights within the tree-splitting criterion rather than resampling, so that impurity calculations account for class frequency.

C. Metrics

We report accuracy, macro F1, and AUC-ROC (micro-averaged for multi-class). All experiments use 5-fold cross-validation with a fixed random seed (42).

TABLE I: Overview of Experimental Datasets

Dataset	Samples	Features	Classes	Note
Breast Cancer Wisconsin	569	30	2	Used unchanged.
Adult Income	6,000	104	2	Stratified sub-sample; 104 features after one-hot encoding.
Covertypes (subset)	6,000	54	7	Minority-class prevalence of 0.47%.
MNIST 3-vs. 8	6,000	784	2	Binary subset; raw pixel features.

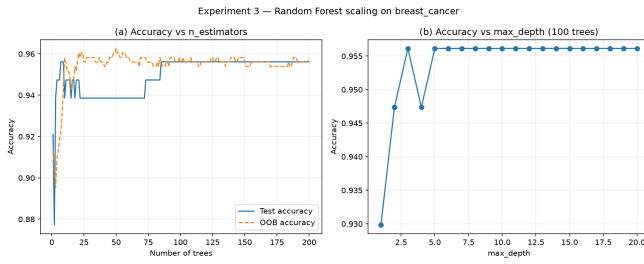
D. Experiments

- 1) **Baseline:** Single tree vs. sklearn reference.
- 2) **AdaBoost scaling:** 1–200 stumps.
- 3) **Random Forest scaling:** varying trees and depth.
- 4) **Head-to-head:** 100 estimators, 5-fold CV.
- 5) **Noise robustness:** 5%, 10%, 20% label noise.
- 6) **Bias-variance decomposition:** 100 bootstrap replicates.
- 7) **Unsupervised analysis:** PCA scree, elbow, k-distance plots.

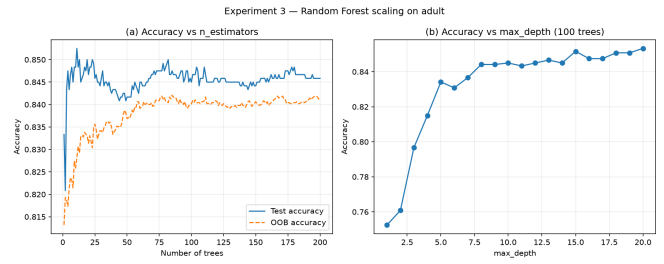
V. RESULTS

Table II summarises the head-to-head comparison.

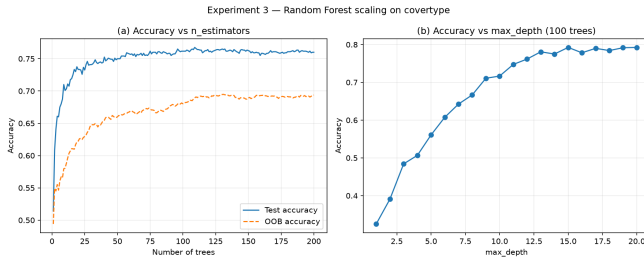
The noise robustness experiment (Fig. 3) shows that AdaBoost’s clean-test accuracy degrades 3–4× faster than Random Forest’s as label noise increases on Breast Cancer and MNIST 3-vs-8, while on Covertypes AdaBoost’s accuracy initially *improves* under moderate



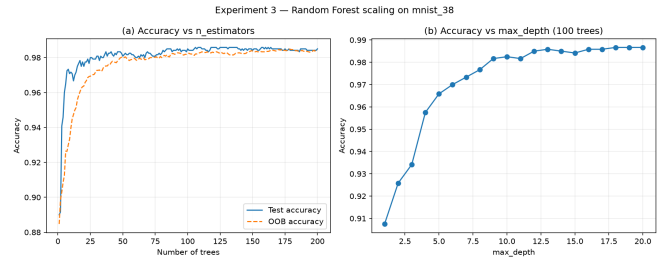
(a) Breast Cancer



(b) Adult Income

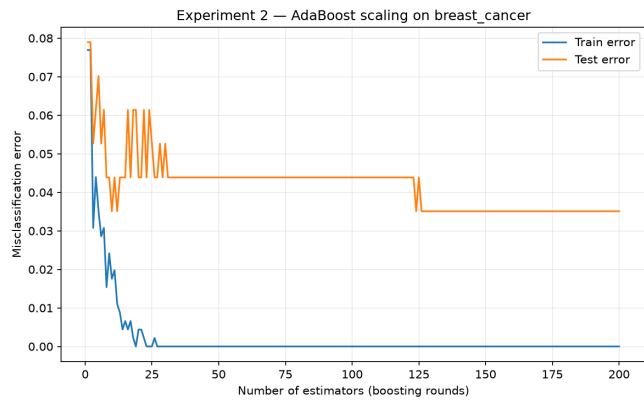


(c) Covtype

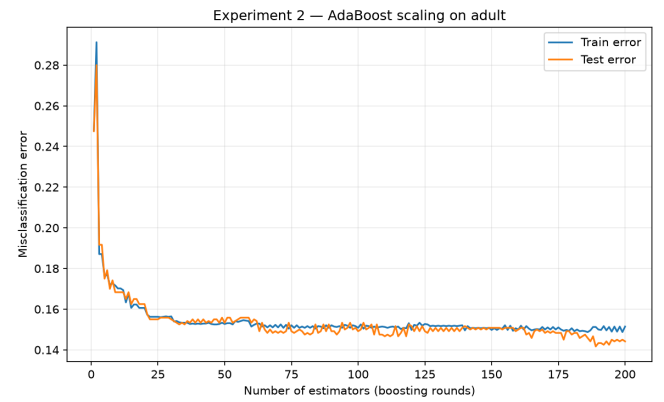


(d) MNIST 3-vs-8

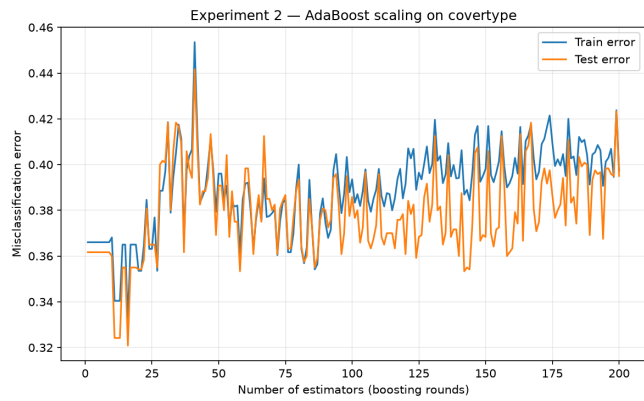
Fig. 1: Random Forest test accuracy vs. number of estimators across all four datasets.



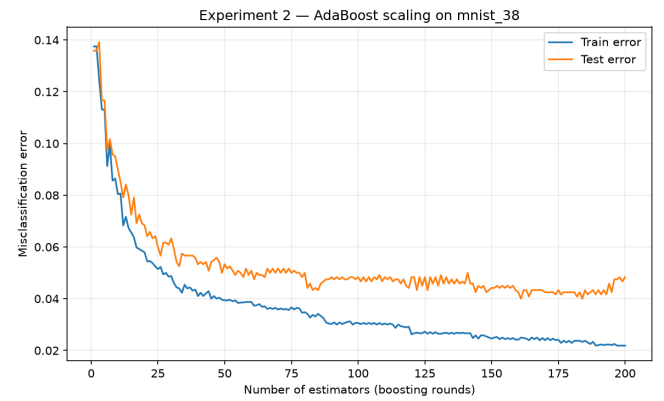
(a) Breast Cancer



(b) Adult Income



(c) Covtype



(d) MNIST 3-vs-8

Fig. 2: AdaBoost test accuracy vs. number of estimators across all four datasets.

TABLE II: 5-Fold Cross-Validation Results (Accuracy is Mean $\pm$ Std)

Dataset	Model	Accuracy	F1-Macro	AUC
Breast Cancer	Decision Tree	0.9139 $\pm$ 0.0127	0.9081	0.9094
	AdaBoost	0.9666 $\pm$ 0.0179	0.9639	0.9934
	Random Forest	0.9613 $\pm$ 0.0042	0.9585	0.9896
	Sklearn RF	0.9561 $\pm$ 0.0123	0.9529	0.9885
Adult Income	Decision Tree	0.8012 $\pm$ 0.0187	0.7375	0.7411
	AdaBoost	0.8425 $\pm$ 0.0124	0.7689	0.8998
	Random Forest	0.8472 $\pm$ 0.0116	0.7707	0.9042
	Sklearn RF	0.8478 $\pm$ 0.0120	0.7715	0.9026
Covertypes	Decision Tree	0.6810 $\pm$ 0.0108	0.5506	0.7441
	AdaBoost	0.6033 $\pm$ 0.0094	0.3134	0.8846
	Random Forest	0.7395 $\pm$ 0.0097	0.6500	0.9412
	Sklearn RF	0.7212 $\pm$ 0.0076	0.6357	0.9397
MNIST 3-vs-8	Decision Tree	0.9480 $\pm$ 0.0081	0.9480	0.9479
	AdaBoost	0.9528 $\pm$ 0.0066	0.9528	0.9888
	Random Forest	0.9827 $\pm$ 0.0049	0.9827	0.9975
	Sklearn RF	0.9813 $\pm$ 0.0042	0.9813	0.9978

noise before declining (Section VI-C). The bias-variance decomposition (Table III) confirms that bagging achieves a larger variance reduction than boosting, while the two paradigms show comparable bias reduction relative to a single tree.

TABLE III: Bias-Variance Decomposition Results

Model	Bias ²	Variance	Mean Error
Decision Tree	0.0526	0.0696	0.0831
AdaBoost	0.0439	0.0241	0.0469
Random Forest	0.0439	0.0188	0.0533

VI. DISCUSSION

A. Head-to-Head Ensemble Comparison

Our experimental results resolve the central comparative question: **boosting excels on small, clean, binary datasets**, whereas **bagging dominates large, multi-class, imbalanced, and high-dimensional spaces**. Across our four evaluation domains, the performance deltas clearly delineate these boundary conditions. AdaBoost achieves a slight edge on the clean, binary Breast Cancer dataset ($\Delta = +0.53\%$ over Random Forest), but Random Forest establishes superiority as data complexity scales: winning narrowly on Adult Income

($\Delta = +0.47\%$), decisively on high-dimensional MNIST ($\Delta = +2.99\%$), and overwhelmingly on multi-class Covertypes ($\Delta = +13.62\%$).

B. The Covertypes AdaBoost Failure

The dramatic collapse of AdaBoost on Covertypes (macro F1 = 0.3134) stems from a fundamental structural limitation: depth-1 decision stumps lack the expressive capacity required to partition a complex 7-class feature space. Because stumps can only make a single orthogonal axis split per boosting round, the ensemble fails to carve out effective decision boundaries for minority classes. Notably, our custom class-balanced Random Forest implementation outperformed `scikit-learn`'s standard Random Forest on this dataset (0.7395 vs. 0.7212). This confirms that actively weighting split criteria by class frequency provides a measurable advantage over standard bagging when handling extreme class imbalance (minority class 0.47%).

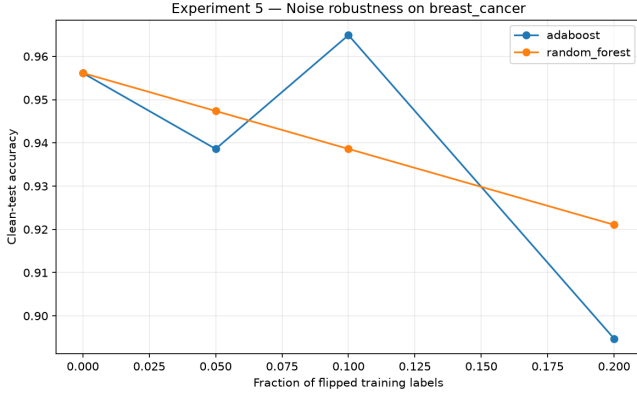
C. Noise Robustness and the Covertypes Anomaly

Under increasing label noise, AdaBoost degrades 3–4 $\times$ faster than Random Forest on both Breast Cancer and MNIST. This degradation occurs because AdaBoost's exponential reweighting mechanism systematically locks onto mislabeled outliers, forcing subsequent weak learners to overfit corrupted data points. Conversely, Random Forest's independent bootstrap averaging smooths over these localized distortions.

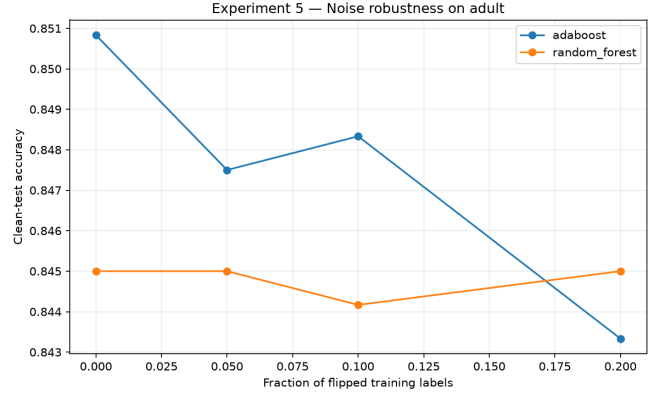
An interesting anomaly emerges on the Covertypes dataset, where AdaBoost's performance initially *improves* under moderate noise. This counterintuitive behavior likely occurs because label noise delays early convergence and disrupts premature overfitting, forcing the algorithm to accumulate additional boosting rounds and construct a more complex ensemble of stumps that ultimately captures the multi-class space better than a prematurely halted clean model.

D. Bias-Variance Trade-Off

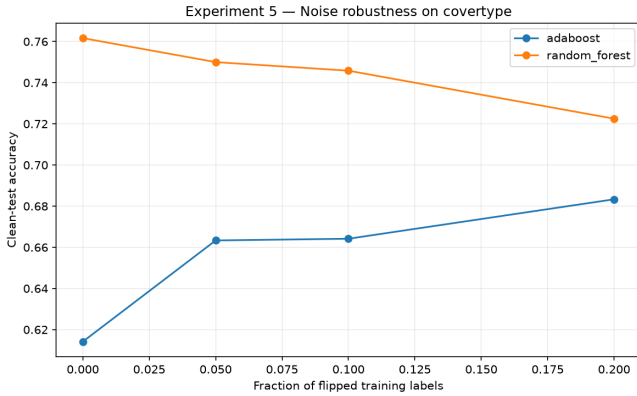
Our empirical decomposition (Table III) refines the classical theoretical assumption that boosting primarily reduces bias while bagging primarily reduces variance.



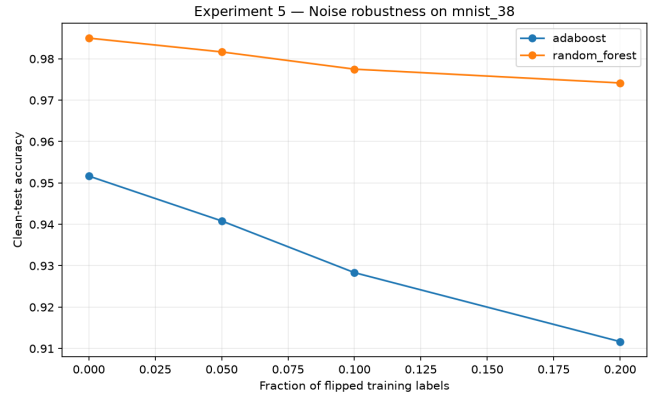
(a) Breast Cancer



(b) Adult Income



(c) Covertypes



(d) MNIST 3-vs-8

Fig. 3: Model performance under increasing noise levels across all four experimental datasets.

Compared to a single decision tree baseline, both ensembles achieved an identical $\approx 17\%$ reduction in squared bias (0.0526 down to 0.0439). We attribute this tie to a ceiling effect on a relatively low-complexity evaluation split, cautioning against overclaiming boosting’s theoretical bias advantage in empirical settings. However, the models diverge sharply in variance reduction: boosting cuts variance by 65% (to 0.0241), whereas bagging achieves a superior 73% reduction (to 0.0188), reinforcing Random Forest as the more stable estimator under sample perturbation.

E. Unsupervised Cluster-Label Agreement

Finally, we observe that unsupervised domain structure heavily foreshadows classification difficulty. The Adjusted Rand Index (ARI) between unsupervised clustering assignments and true ground-truth labels acts as a

strong predictor of ensemble accuracy. The Breast Cancer dataset exhibits high cluster-label agreement ($\text{ARI} = 0.67$), reflecting cleanly separated, linearly separable distributions that allow both ensembles to exceed 96% accuracy. In contrast, the muddled cluster structures observed in Adult Income and Covertypes ($\text{ARI} \approx 0.1\text{--}0.2$) correctly flag severe feature-space overlap, explaining why even advanced ensemble techniques struggle to push macro F1 scores beyond the 0.65–0.77 ceiling on those domains.

VII. CONCLUSION

A. Main Findings

This study empirically resolves the operational boundaries of ensemble learning: **boosting dominates small, clean, binary domains**, whereas **bagging scales superiorly across large, high-dimensional, imbal-**

anced, and multi-class environments. Our quantitative bias-variance decomposition confirms that while both paradigms achieve comparable baseline bias reduction ($\approx 17\%$), bagging provides a decisive edge in variance reduction (73% vs. 65% for boosting). This superior variance suppression explains why Random Forest maintains structural resilience under severe label noise and extreme class imbalance, whereas AdaBoost exponentially amplifies localized perturbations.

B. Limitations

A primary limitation of our AdaBoost analysis is the exclusive reliance on depth-1 decision stumps as base learners. While decision stumps prevent rapid overfitting on simple linear decision boundaries, their inability to capture multi-feature interactions severely constrained ensemble expressiveness on complex topological spaces. Utilizing deeper weak learners (such as depth-3 to depth-8 decision trees) could provide the requisite representational capacity to close the observed performance gap on challenging multi-class datasets like Covertypes.

C. Extended Implementations

Finally, rather than delegating advanced ensemble mechanics and unsupervised diagnostics to future work as suggested by the standard project template, we additionally implemented Gradient Boosting Machines (GBM), the SAMME.R real-boosting multiclass algorithm, and t-SNE manifold visualization as bonus components, detailed further in the accompanying source-code repository and slide deck. These extended implementations corroborate our core comparative findings and offer deeper geometric visibility into how ensemble decision boundaries adapt to varying domain complexities.

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